

CHEM 515 Practice Session 2: Density Functional Theory Calculations

Name: _____

In this practice, you will use IQmol to set up density functional theory (DFT) calculations for Q-Chem. You will use those calculations to investigate (i) intermolecular interactions and (ii) the ring strain of cycloalkanes. You will also gain experience in (i) running calculations on the Itcotck server and (ii) using the ORCA package, if these are new to you.

You will need to **complete the tables and short-answer questions on this handout** and also generate the requested files for submission. Please refer to the **Assignment page** on Canvas for detailed submission guidelines. The grading will be based on attendance (2 pts) and your submitted work (8 pts).

Exercise 1: Calculating intermolecular interactions using DFT

In this exercise, you will

- Compare the accuracy of different density functionals in predicting the water dimer interaction energy
- Compare the H-bond strengths in H_2O , NH_3 , and HF dimers
- Assess the effectiveness of counterpoise corrections for BSSE for different basis sets
- Generate the potential energy curve for the dissociation of water dimer

Note: for this exercise, please (i) Do NOT optimize the structures; (ii) Set `SCF_CONVERGENCE` = 8; (iii) Set `GUI` = 0 exclusively.

Interaction energy for the water dimer

Your first task is to calculate the interaction energy between two water molecules at the equilibrium geometry of the water dimer (provided) using the functionals listed in **Table 1** and the def2-TZVPD basis set. The equilibrium structure and the reference value for the interaction energy are obtained from the [A24 dataset](#) (*J. Chem. Theory Comput.* **2013**, 9, 2151–2155). **Steps to follow:**

- Open “water_dimer.eq.xyz” and go to “Q-Chem Setup”. Set “Method” to “Omega-B97X-V” (which is $\omega\text{B97X-V}$), and then manually type “BASIS = def2-TZVPD” in the generated input. Remember to change “Convergence” (`SCF_CONVERGENCE`) to 8 and `GUI` = 0.
- To speed up the process of running multiple jobs, here we take advantage of the feature of running multiple Q-Chem jobs *sequentially*. After you finish setting up the first job, **click the green “+”** next to “Edit”. You will see “Job 2” is now after “Job Section” and that a new input shows up which is separated from the original input (now faded) by “@@@”. You now just need to change the “Method” for the 2nd job to “M06-2X”. Make sure that the other settings are the same as the 1st job.

The 3rd and 4th jobs can be added in the same way: for B3LYP-D3(BJ) and B3LYP-D3(0), you should first select “B3LYP” as the “Method”, and then go to “Advanced → DFT → Dispersion Correction” and set the Model to “DFT-D3(BJ)” and “DFT-D3(0)”, respectively.

Submit this job to the Q-Chem server. The calculation should be able to finish in about 1.5 minutes. Once you copy the job back from the server, you will see **4 numbers** under “Geometries”, which corresponds to the results of 4 different functionals.

- Now you need to calculate the energy of each monomer using all 4 functionals. Re-select “water.dimer.eq” under “Model View” and go to “Q-Chem Setup” again. Click “Reset” on the right bottom of the control panel to clean things up. You can then delete the *last* 3 rows of the XYZ coordinates under the \$molecule section (the remaining coordinates correspond to **monomer 1**), and then you can set up and run the 4 consecutive jobs in the same way as above.
- Repeat the above for the 2nd water monomer (now delete the first 3 rows of XYZ coordinates under the \$molecule section) and report all the monomer energies in **Table 1**. You can now evaluate the intermolecular interaction energy (ΔE_{int}) using

$$\Delta E_{\text{INT}} = E(\text{dimer}) - E(\text{monomer 1}) - E(\text{monomer 2}) \quad (1)$$

Note: 1 a.u. (hartree) = 627.5095 kcal/mol; you may want to calculate this on Excel.

Table 1: Evaluation of the water dimer interaction energy using different functionals. All energies should be reported in a.u. (with 6 digits after decimal point) except for the last column. **(1 pt)**

Functional	dimer	monomer 1	monomer 2	ΔE_{INT} (kcal/mol)
ω B97X-V				
M06-2X				
B3LYP-D3(BJ)				
B3LYP-D3(0)				

Questions:

1. The reference value for the water dimer interaction energy (calculated at the CCSD(T)/CBS level) is -5.006 kcal/mol. Compared to this reference, which functional performs the best with the def2-TZVPD basis? What are the absolute (in kcal/mol) and relative (in %) errors? **(0.25 pt)**
2. Which flavor of “-D3” correction gives a more accurate interaction energy for the water dimer when paired with the B3LYP functional? Comparing the HOMO energies calculated by B3LYP-D3(0) and B3LYP-D3(BJ), would you expect to see any difference? Explain your answer. **(0.25 pt)**
3. The M06-2X functional is *not* dispersion-corrected. Based on the result in Table 1, do you think adding dispersion correction will improve its accuracy for the water dimer? Explain your answer. **(0.25 pt)**

Comparison of hydrogen bond strengths

Now we compare the strength of the hydrogen bond in water dimer to those in the HF dimer and NH_3 dimer: with the given equilibrium structures (“HF_dimer_eq.xyz” and “NH3_dimer_eq.xyz”), calculate the interaction energies for these two dimer complexes at the $\omega\text{B97M-V}/\text{def2-QZVPD}$ level and report the results in **Table 2**. Please note that the requested functional here has *changed*: you need to manually type “METHOD = ωB97MV ” for $\omega\text{B97M-V}$.

For these calculations, you should (i) set up the dimer calculation and two monomer calculations for each system as 3 jobs in tandem; (ii) execute these calculations on the Itcotck server through Linux command lines. The command to run Q-Chem calculations on the Itcotck server:

```
qchem -nt 4 [job.in] [job.out] &
```

Once the jobs are done, you can execute a simple Linux command under the same directory to extract the 3 energies in each job: e.g.,

```
grep "Total energy" water_dimer_wb97mv_qzvdp.out
```

for the water dimer system. You can also open the output using IQmol (ignoring the warning message), and search for “Total energy” in the output text.

Table 2: Interaction energies for the three H-bond dimers calculated at the $\omega\text{B97M-V}/\text{def2-QZVPD}$ level (0.5 pt)

Species	dimer	monomer 1	monomer 2	ΔE_{INT} (kcal/mol)	Ref (kcal/mol)
H_2O dimer					
HF dimer					
NH_3 dimer					

Questions:

1. Based on the magnitude of the interactions, which system has the *weakest* hydrogen bond? (0.2 pt)
2. Complete the last column of the table by looking up the CCSD(T)/CBS on the website for the A24 dataset ([link](#)). Then calculate the root-mean-square error (RMSE) of $\omega\text{B97M-V}/\text{def2-QZVPD}$ on these three interactions. (0.3 pt)

Assessment of BSSE and the effectiveness of counterpoise correction

With basis sets that are smaller than def2-TZVPD, counterpoise (CP) corrections for BSSE needs to be applied to obtain accurate results. As shown in **lecture 7**, Q-Chem offers a convenient job control option for that purpose with “JOBTYPE = BSSE”. Follow the procedure below to set up interaction energy calculations with CP correction using $\omega\text{B97X-V}$ and *three* different basis sets: 6-31+G(d), def2-SVPD, and def2-TZVP:

- Reopen “water_dimer_eq.xyz” using IQmol. For “BSSE” calculations, we need to specify fragments (each fragment corresponds to a monomer) in the \$molecule section: after specifying the charge and multiplicity for the whole molecule, each fragment should start with “—” and then each fragment’s charge and multiplicity need to be specified. For our water dimer system, the “\$molecule” section should look like that in **Fig. 1**.

```

$molecule
0 1
--
0 1
O   -0.0669991    0.0000000    1.4943547
H    0.8157343    0.0000000    1.8658664
H    0.0688551    0.0000000    0.5391428
--
0 1
O    0.0625477    0.0000000   -1.4226321
H   -0.4069654   -0.7601784   -1.7717445
H   -0.4069654    0.7601784   -1.7717445
$end

```

Figure 1: The \$molecule section for the water dimer separated into two fragments.

- Under “Calculate”, select “BSSE Correction”. The functional and basis set can be set as before (note: 6-31+G(d) $\equiv$ 6-31+G*). Remember to change “SCF_CONVERGENCE” to 8 and set “GUI = 0”.
- Submit the job to the Q-Chem server (you may also run the job on the Ictotck server). Once finished, open the output file using IQmol (expand the job under Model View and then expand “Files”). Find the number of basis functions first (by searching for “basis functions” in the output) and then scroll down to the bottom. The numbers under “E(CP-CORR)” are CP-corrected monomer energies, which should be reported in **Table 3**, and so is the energy of the “Complex” (dimer). The final interaction energies can still be calculated using Eq. (1), where the monomer energies should be the CP-corrected ones.

Table 3: Evaluation of water dimer interaction energies using smaller basis sets and counterpoise corrections for BSSE (**0.75 pt**)

Basis set	# of Basis	monomer 1 (CP)	monomer 2 (CP)	dimer	ΔE_{INT} (kcal/mol)
6-31+G(d)					
def2-SVPD					
def2-TZVP					

Question:

1. Which basis set yields the best agreement with the reference value (-5.006 kcal/mol) once the CP correction is applied? Does it coincide with the basis set that has greatest number of basis functions. (**0.3 pt**)
2. How does the accuracy of the best-performing basis set in Table 3 compare to that of ω B97X-V/def2-TZVPD without counterpoise correction? (**0.2 pt**)

The potential energy curve of water dimer

We are now going to generate the potential energy curve for the dissociation of the water dimer at the ω B97X-D3/def2-SVPD level (Note: in reality you should consider using an even larger basis such as def2-TZVPD)

for this task). This can be done with the following steps:

- Reopen “water_dimer_eq.xyz”. Under the “Select Mode” (the magic wand) click the two oxygen atoms on each water molecules. Then select “Build → Set Geometric Constraint”. A small window should pop up.
- Select “Scan”, and then set the starting and ending distances to 2.6 and 3.6 Å, respectively; set the number of points to 11 so that the increment will be 0.1 Å. Click “Apply”. You should see a purple line connecting two O atoms with a little “disk” in the middle.
- Now go to “Q-Chem Setup”. Q-Chem has a job type called “PES scan” that is specifically designed for this type of task. Select that under “Calculate”. Now you should see a small section appear in the generated input:

```
$scan
stre 1 4 2.600 3.600 0.100
$end
```

This indicates the scan for the distance (“stretch”) between atoms 1 and 4 from 2.6 to 3.6 Å with an increment of 0.1 Å.

- Manually set the “Method” to “wB97X-D3” and “Basis” to “def2-SVPD”. Also set SCF_CONVERGENCE to 8 and “GUI” to 0. **You should run this calculation on the Itcotck server.** This job may take a few minutes to finish.
- Copy the output back once the job is finished and open it with IQmol. Expand the job under “Model View” and then double-click on “Scan Geometries”. You should see a plot of the potential energy curve. Right-click on the plot and save the image, **which will need to be submitted.** (0.25 pt)

Question: Under which O···O distance does the water dimer have the most favorable energy? Is it in good agreement with the O···O distance in the initial equilibrium structure (“water_dimer_eq.xyz”)? (0.25 pt)

Note: Before you start Exercise 2, you may want to clean up the list under “Calculation → Job Monitor”, and probably restart IQmol.

Additional exercise: Assessment of r2SCAN-3c for water dimer interaction

r2SCAN-3c (*J. Chem. Phys.* **2021**, 154, 064103) is a composite electronic structure method (similar to the PBEh-3c method used in Exercise 2). It shows excellent performance for many chemical applications. This method hasn’t been implemented in Q-Chem yet, but it is available in ORCA. Here we are going to assess the accuracy of this method for the interaction energy of the H₂O dimer.

The ORCA input file (“water_dimer_r2scan3c.inp”) has been provided on Canvas. It looks pretty simple since we are just using ORCA’s default SCF settings. Also, r2SCAN-3c is supposed to be only used with one basis set called def2-mTZVPP (a pruned def2-TZVPP basis), which will be automatically invoked.

To run this ORCA calculation, you first need to set up the environment for ORCA, which can be done by executing `source /home/ymao/scripts/setup-orca.sh` (you only need to do this once for each new terminal window), then do

```
$orcadir/orca [job.inp] > [job.out] &
```

After completing the dimer calculation, you can modify the input yourself and create the input files for the two monomers (just by deleting a few lines in the XYZ coordinates). After you complete all three calculations, you can extract the results for all 3 jobs by executing

```
grep "Total Energy" *.out
```

Table 4: Water dimer interaction energy calculated using the r2SCAN-3c method

dimer	monomer 1	monomer 2	ΔE_{INT} (kcal/mol)

Complete **Table 4** based on the results. **(0.3 pt)** Then calculate the absolute (in kcal/mol) and relative (%) error compared against the reference value. **(0.2 pt)**

Exercise 2: Quantifying ring strain in cycloalkanes

(Note: This exercise is adapted from the “[Predicting Trends in Ring Strain of Cycloalkanes](#)” lab on the Q-Chem website)

In this exercise, you will use a low-cost DFT method PBEh-3c (*J. Chem. Phys.* **2015**, 143, 054107) to predict the ring strain in cycloalkanes C_nH_{2n} with $n = 3-8$. The cyclic alkanes can be considered as rings of CH_2 groups. If there is no ring strain, then the energy of a cycloalkane C_nH_{2n} should simply be n times the energy of a CH_2 group in an *infinitely long straight chain alkane*. We thus define the ring strain as the following:

$$E_{\text{RS}} = E[\text{C}_n\text{H}_{2n}] - nE[\text{CH}_2] \quad (2)$$

where E_{RS} is the ring-strain energy, $E[\text{C}_n\text{H}_{2n}]$ is the energy of a cycloalkane with n carbon atoms (e.g., $n = 3$ for cyclopropane, $n = 4$ for cyclobutane, etc.), and $E[\text{CH}_2]$ is the energy of a CH_2 group in an infinite straight chain alkane.

Let’s determine the value of $E[\text{CH}_2]$ first with the following procedure:

1. Draw the propane molecule using IQmol with the procedure as you constructed pyridine in our last practice (Build the carbon skeleton → Add hydrogens → click the “E↓” button to minimize the energy)
2. We are now going to perform a PBEh-3c calculation directly with this structure *without* further geometry optimization. Open “Q-Chem Setup”. Make sure that you are calculating “Energy”. The PBEh-3c method is not supported by IQmol yet so you need to specify the following manually: “METHOD = PBEh-3c”, “BASIS = def2-mSVP” (Note: PBEh-3c should only be used with this special basis), and then remember to set “Convergence” to 8 and “GUI = 0”.
3. Submit the calculation to the server. When it is done, report the energy in the 2nd column (“ $E(\text{chain})$ ”) of **Table 5**.

Repeat steps 1–3 for straight chain alkanes with $n = 4-8$ and then further evaluate the results for the 3rd column: $E(n) = E(\text{chain})/n$ (you may want to calculate them using **Excel**). **(0.5 pt)**

In order to obtain the value of $E(n)$ at $n \rightarrow \infty$, which will be the value of $E[\text{CH}_2]$ that we need, we can assume that $E(n)$ depends on n as:

$$E(n) = a + b/n \quad (3)$$

You can obtain the parameters a and b from *least-square fitting*. An easy way to do so is to use Excel to generate a scatter plot of $E(n)$ vs. $1/n$, and then use the “Add Trendline” feature. Make sure to display the fitted equation and the R^2 value on the chart. If you did so, change the title of that plot to “ $E(n)$ vs. $1/n$ ” in Excel and then save the figure, **which needs to be submitted with your report (0.25 pt)**. You can use other software to do the least-square fitting and generate a plot that show *the original data and the fitted line*. Report the values of fitted parameters a and b as well as the R^2 value below **(0.25 pt)**:

Table 5: Determination of the ring strain in cycloalkanes. The ring-strain energy (E_{RS}) should be reported in kcal/mol with 2 digits after decimal point, and the rest are in a.u. (6 digits required).

n	$E(\text{chain})$ (a.u.)	$E(\text{chain})/n$ (a.u.)	$E(\text{cyclo})$ (a.u.)	E_{RS} (kcal/mol)
3				
4				
5				
6				
7				
8				

Based on the R^2 value you obtained, does Eq. (3) provide a good description for the relationship between $E(n)$ and n ? **(0.25 pt)**

Regardless of your answer to the previous question, we will proceed with using the value of the fitted parameter a for $E[\text{CH}_2]$. Now, repeat the steps 1–3 above for cycloalkanes with $n = 3$ –8. Report your results in the 4th column of **Table 5 (0.5 pt)**, and then calculate the values of ring-strain energy E_{RS} for each of the cycloalkanes based on Eq. (2) and complete the last column of the table. **(0.5 pt)**

Using Excel or other software, **generate a plot of E_{RS} vs. the number of carbon atoms (which also needs to be submitted).** **(0.25 pt)** Discuss the trend you observe on this plot and rationalize the results based on what you learned in organic chemistry classes **(0.5 pt)**.

Hint: you can check the typical C–C–C bond angles in each cycloalkane using IQmol and compare them to the ideal bond angle for sp^3 carbon.